BARODA UNIVERSITY
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"Think Big... Think Beyond"

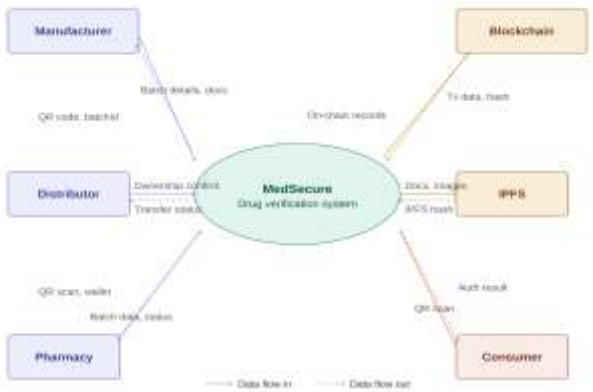
MEDSECURE: BLOCKCHAIN & AI-POWERED COUNTERFEIT MEDICINE DETECTION

Brief Description:

MedSecure is a final-year B.Tech project that tackles the growing problem of fake medicines in the supply chain. It uses Ethereum blockchain to permanently record every medicine batch from the moment it is made, through distributors and pharmacies, all the way to the consumer. Every transfer of ownership is locked on-chain — meaning nobody can alter the history. A computer vision module (powered by OpenCV) compares live packaging photos against the original manufacturer images to detect counterfeits. Consumers simply scan a QR code on the medicine box to instantly verify whether it is genuine — no technical knowledge required.

Principle:

The system's core relies on Ethereum smart contracts that enforce strict rules — only the current owner of a medicine batch can transfer it to the next party. Medicine data and packaging images are stored on IPFS (a decentralised file system), with only the reference hash written on-chain. A QR code is generated for each batch; once a unit is sold, the QR is permanently locked to prevent re-use by counterfeiters. The computer vision module uses ORB feature matching and SSIM comparison to produce an authenticity confidence score — above 75% means genuine.



Beneficiaries:

- Consumers buying medicines
- Pharmacies & retail chemists
- Medicine manufacturers
- Distributors & wholesalers
- Drug regulatory authorities
- Healthcare organisations

Benefits:

- Protects patients from counterfeit drugs
- Full, tamper-proof supply chain history
- Instant QR-based verification for anyone
- AI-powered packaging fraud detection
- No single point of failure (decentralised)
- Helps regulators trace recalled medicines

Technologies Used:

- **Ethereum & Solidity**
Smart contracts on Sepolia testnet via Foundry
- **IPFS**
Decentralised document & image storage
- **Next.js + Viem + Wagmi**
Web frontend with MetaMask wallet integration
- **OpenCV (Python)**
ORB feature matching + SSIM for packaging auth
- **PostgreSQL**
Off-chain audit logs, QR scan & session records
- **Solidity Smart Contract**
On-chain role-based access, lifecycle state machine

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